gdTAI V4.2 Step 0

gdTAI V4.2 Step 0: why V4.1 failed and how to repair it

Decision summary

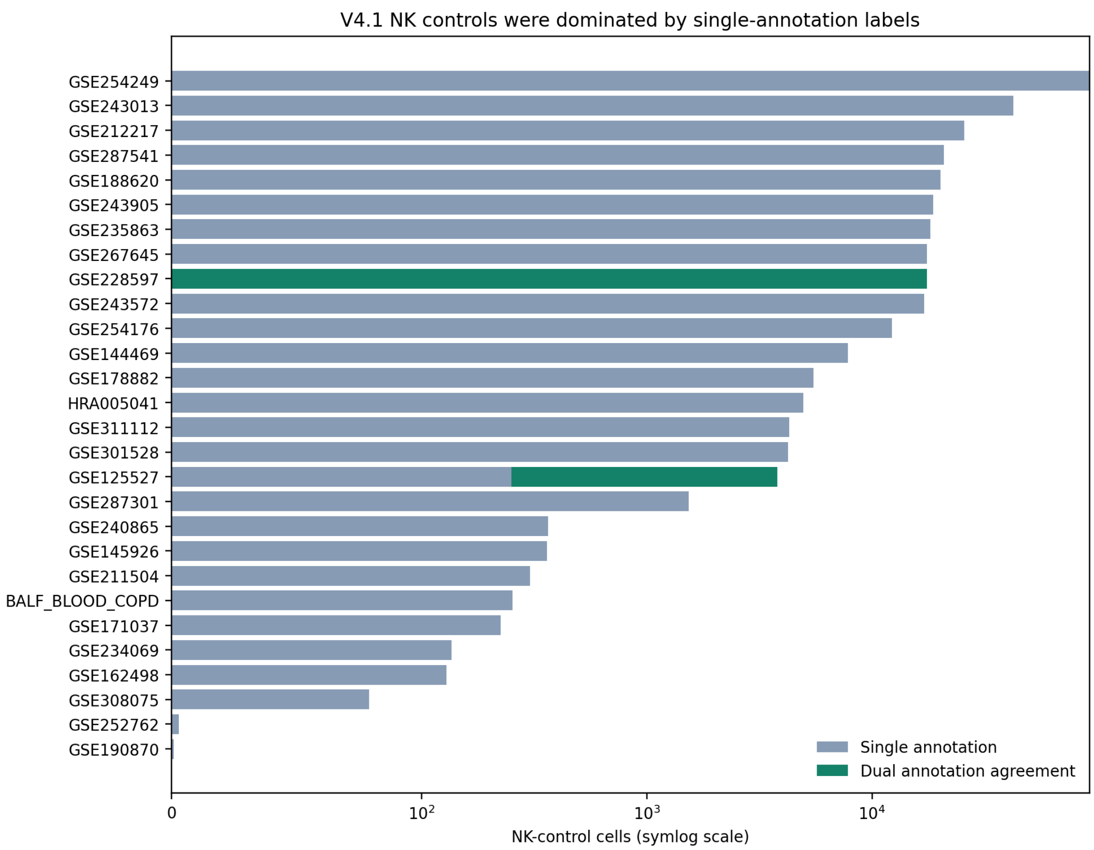
V4.1 did **not** demonstrate that a two-step gdTAI model is intrinsically worse than V2 or V3. It stopped before Stage 2 because the Stage-1 eligibility gate treated a heterogeneous, mostly weak NK-control pool as equally definitive ground truth.

- V4.1 used **336,780** strict-NK controls.
- Only **21,054 (6.3%)** had dual scVI/author NK agreement.
- **315,726 (93.7%)** had only the scVI NK label after conflict filtering.
- Under the unchanged T-recall thresholds, the worst source with at least 100 cells passed at **91.0% to 99.2%** for all V4.1 controls.
- The same saved out-of-fold predictions passed only **5.3% to 19.4%** of dual-annotation NK controls. All 15 candidate/fold combinations satisfy the frozen 50% cap under that stronger truth definition.

This is a **label-only counterfactual**, not V4.2 performance. The V4.1 models were still trained using the weak controls, and no model was fitted in this audit.

Why V4.1 looked weak

1. **Stage 2 never ran.** There is no V4.1 final F1, precision, or recall to compare directly with V2/V3.
2. **The Stage-1 failure was source-specific.** Large single-annotation sources such as GSE144469 and GSE287301 were highly T-like by CD3/TCR-constant expression, while genuine-looking BALF NK controls also showed substantial passage in one outer fold. This combines label uncertainty with real cross-dataset calibration shift.
3. **The gate was brittle.** It used the literal maximum source passage, including a one-cell source, and did not distinguish primary labels from stress labels.
4. **Historical V2/V3 figures are optimistic for model selection.** Their validation cohorts were reused during development. V4.2 must compare algorithms on identical grouped, fold-local splits and report the frozen historical artifacts separately.

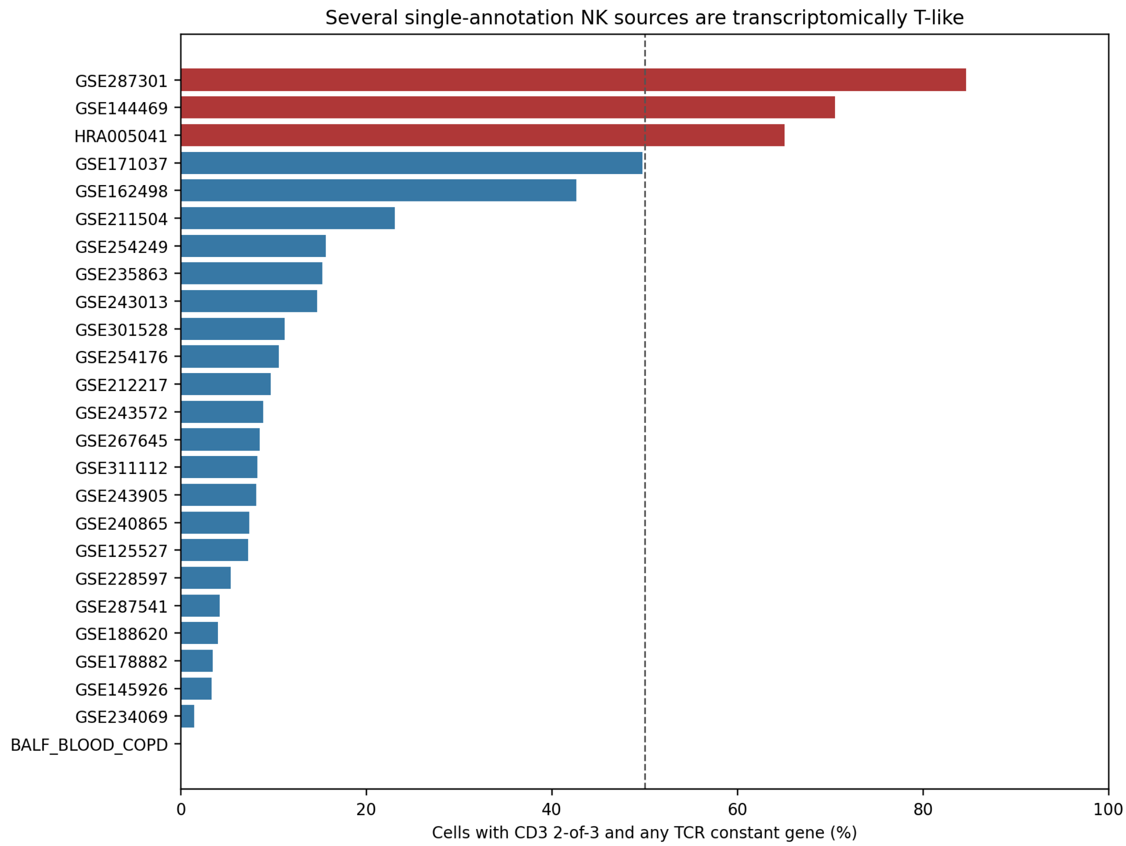


NK-control provenance

NK-label audit

The expression audit is used only to diagnose label conflict. It does not create, remove, or relabel training truth.

source_gse_id	n_cells	cd3_two_of_three_fraction	expression_coherent_t_fraction	ab_cd3_coherent_fraction	gd_cd3_coherent_fraction
GSE287301	1539	90.3%	84.7%	80.7%	53.5%
GSE144469	7804	73.0%	70.5%	69.9%	23.4%
HRA005041	4951	69.1%	65.1%	61.5%	44.0%
GSE171037	225	49.8%	49.8%	49.8%	14.2%
GSE162498	129	42.6%	42.6%	38.0%	33.3%
GSE211504	303	26.7%	23.1%	21.1%	12.5%

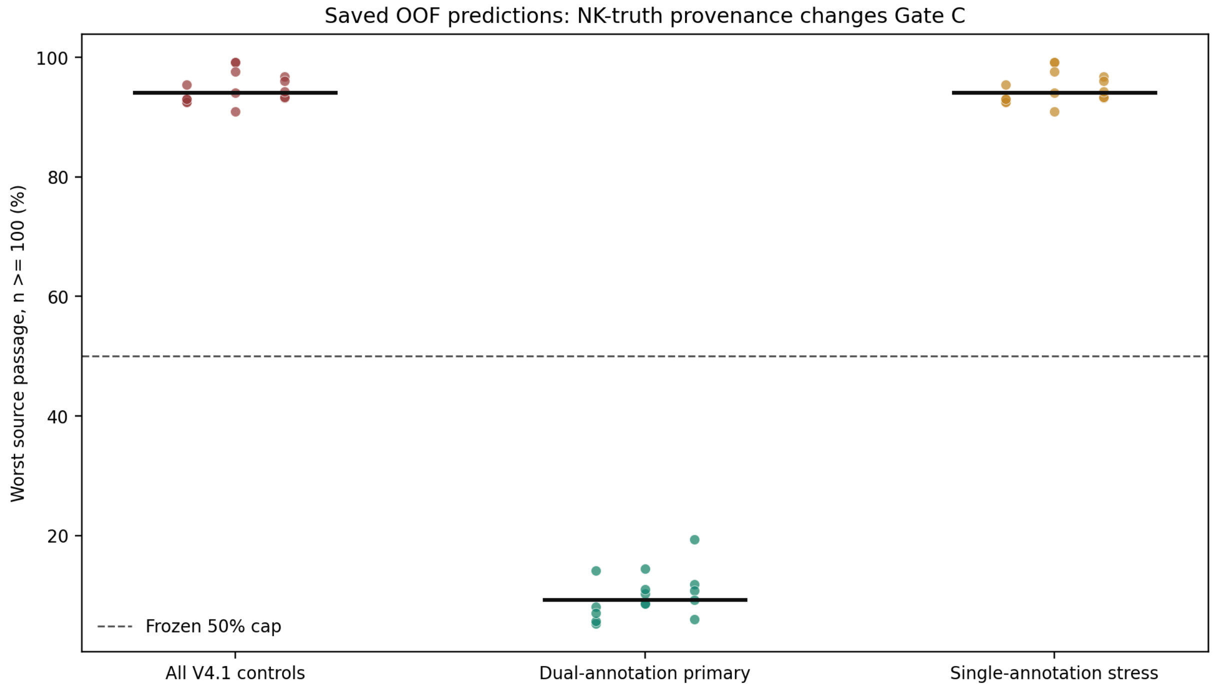


T-lineage coherence

Saved-OOF counterfactual

For each of the 15 completed V4.1 Stage-1 candidate/fold combinations, the audit reconstructed the exact checksum-bound training rows, loaded calibrated OOF probabilities, and selected the highest threshold satisfying per-source gdT recall >= 0.99 and productive alpha-beta T-cell recall >= 0.98. Only the NK evaluation mask changed.

heldout_source	family	threshold	minimum_source_gdt_recall	minimum_source_abt_recall	pooled_passage	maximum_source_passage_n_ge_minimum	worst_source_n_ge_minimum
BALF_BLOOD_COPD	torch_ridge	0.00257	99.02%	99.92%	2.92%	5.92%	GSE125527
GSE144469	torch_ridge	0.00043	99.02%	99.84%	4.88%	8.60%	GSE125527
HRA005041	torch_ridge	0.00126	99.03%	99.41%	4.95%	5.27%	GSE125527



Counterfactual feasibility

Frozen V4.2 repair

The detailed precommit is in `docs/GDTAI_V4_2_PRECOMMITTED_PLAN.md`. Its key changes are:

- Primary Stage-1 NK truth requires dual scVI/author NK agreement, no productive TCR, and no doublet flag.
- Single-annotation NK cells receive zero fitting and selection weight; they remain a required stress test.
- CD3/TCR expression coherence remains diagnostic only and cannot define truth.
- Stage 1 remains a high-recall T-cell gate. It cannot use a hard cytotoxic-gene veto, and Stage 2 has no hard TRDC/TRDV requirement.
- Stage 2 uses individual genes and the cross-fitted Stage-1 probability, never TRD/TRAB scores, source IDs, or batch IDs; total features remain below 300.
- Source-size-aware constraints use strata with at least 100 controls. Smaller strata and Wilson intervals are reported separately rather than controlling a literal maximum.
- V2-like and V3-like algorithms are retrained fold-locally on the same labels and splits. Frozen V2/V3 artifacts remain descriptive because their development data overlap the benchmark.
- Promotion requires a clinically meaningful F1 improvement under grouped resampling while retaining the frozen paired-alpha-beta and NK false-positive guardrails.

QC and supervision gate

check	status	detail
sha256_cell_manifest	PASS	8157cbebfedeb84fc34eba05429aa8a7a834c6f7ceba15bd4790b
sha256_feature_manifest	PASS	7fc2b0afb2b761288f941895a219fb38cc541a49464c8d2b330b81
sha256_gene_feature_cache	PASS	97b2a719cfe77526005941fce289e676518513cec7104aa777082
feature_cache_shape	PASS	(1137739, 197)
checkpoint_rows::outer_0_HRA005041::torch_ridge__l2_strength=0.001	PASS	4b2ed0dd323846955ea2219412665cec7158e1eb84160fe53d25
checkpoint_rows::outer_0_HRA005041::xgboost__colsample_bytree=1.0__learning_rate=0.05__max_depth=5__min_child_weight=200__n_estimators=200__reg_lambda=10.0__subsample=1.0	PASS	4b2ed0dd323846955ea2219412665cec7158e1eb84160fe53d25
checkpoint_rows::outer_0_HRA005041::torch_ridge__l2_strength=0.01	PASS	4b2ed0dd323846955ea2219412665cec7158e1eb84160fe53d25
checkpoint_rows::outer_0_HRA005041::xgboost__colsample_bytree=1.0__learning_rate=0.05__max_depth=3__min_child_weight=50__n_estimators=200__reg_lambda=10.0__subsample=1.0	PASS	4b2ed0dd323846955ea2219412665cec7158e1eb84160fe53d25
checkpoint_rows::outer_0_HRA005041::torch_ridge__l2_strength=0.1	PASS	4b2ed0dd323846955ea2219412665cec7158e1eb84160fe53d25
checkpoint_rows::outer_1_GSE144469::torch_ridge__l2_strength=0.001	PASS	49ef83ce2c2cddd8d05032273721e59bcaeacef2c72a41a2aaff59
checkpoint_rows::outer_1_GSE144469::xgboost__colsample_bytree=1.0__learning_rate=0.05__max_depth=5__min_child_weight=200__n_estimators=200__reg_lambda=10.0__subsample=1.0	PASS	49ef83ce2c2cddd8d05032273721e59bcaeacef2c72a41a2aaff59
checkpoint_rows::outer_1_GSE144469::torch_ridge__l2_strength=0.01	PASS	49ef83ce2c2cddd8d05032273721e59bcaeacef2c72a41a2aaff59
checkpoint_rows::outer_1_GSE144469::xgboost__colsample_bytree=1.0__learning_rate=0.05__max_depth=3__min_child_weight=50__n_estimators=200__reg_lambda=10.0__subsample=1.0	PASS	49ef83ce2c2cddd8d05032273721e59bcaeacef2c72a41a2aaff59
checkpoint_rows::outer_1_GSE144469::torch_ridge__l2_strength=0.1	PASS	49ef83ce2c2cddd8d05032273721e59bcaeacef2c72a41a2aaff59
checkpoint_rows::outer_2_BALF_BLOOD_COPD::torch_ridge__l2_strength=0.001	PASS	bad99923cb15679819ad5ee16d01cbbe23eaca9a99e5d0198744
checkpoint_rows::outer_2_BALF_BLOOD_COPD::xgboost__colsample_bytree=1.0__learning_rate=0.05__max_depth=5__min_child_weight=200__n_estimators=200__reg_lambda=10.0__subsample=1.0	PASS	bad99923cb15679819ad5ee16d01cbbe23eaca9a99e5d0198744
checkpoint_rows::outer_2_BALF_BLOOD_COPD::torch_ridge__l2_strength=0.01	PASS	bad99923cb15679819ad5ee16d01cbbe23eaca9a99e5d0198744
checkpoint_rows::outer_2_BALF_BLOOD_COPD::xgboost__colsample_bytree=1.0__learning_rate=0.05__max_depth=3__min_child_weight=50__n_estimators=200__reg_lambda=10.0__subsample=1.0	PASS	bad99923cb15679819ad5ee16d01cbbe23eaca9a99e5d0198744
checkpoint_rows::outer_2_BALF_BLOOD_COPD::torch_ridge__l2_strength=0.1	PASS	bad99923cb15679819ad5ee16d01cbbe23eaca9a99e5d0198744
stage1_checkpoints_complete	PASS	15/15 candidate-fold combinations
dual_annotation_nk_sources	PASS	2 sources
dual_annotation_counterfactual_gate	PASS	15/15 pass Wilson-aware 50% cap
expression_used_for_truth	PASS	diagnostic only; no labels changed
model_fitting_performed	PASS	none; saved V4.1 OOF probabilities only

Result: PASS. This authorizes review of the V4.2 protocol only. It does not authorize fitting. The next action requires explicit supervision approval.

Reproducible outputs

- Full tables: `Integrated_dataset/tables/gdT_prediction/gdtai_v4_2_step0/`
- Machine summary: `Integrated_dataset/logs/gdT_prediction/gdtai_v4_2_step0/summary.json`
- Frozen protocol: `docs/GDTAI_V4_2_PRECOMMITTED_PLAN.md`
- Reader HTML: `gdT_prediction/gdtai_v4_2_step0/index.html`
- PDF: `gdT_prediction/gdtai_v4_2_step0/gdtai_v4_2_step0_report.pdf`